

CP020003 — Artificial Intelligence 2026

Khon Kaen University

Week 7— Homework Assignment

In class this week you built a full association-rule-mining pipeline — Apriori, FP-Growth, and an embedding-based Item2Vec model — on the KKU Online Retail transaction log, restricted to France so the results were small enough to inspect by eye. This assignment is deliberately open-ended: you will use the exact same dataset, but you choose your own market segment, your own min_support strategy, and your own combination of techniques to investigate a question you find genuinely interesting. There is no single "correct" set of rules to find — you are graded on how rigorously you mine, filter, and justify the patterns you report, and how convincingly you turn them into a business recommendation.



1. Dataset

Use the exact same dataset from the in-class notebook — download it directly from the course GitHub repository below and unzip it (it contains a single CSV file):

https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1/blob/main/dataset/KKU-EUOnlineRetail.csv.zip

The CSV has one row per item purchased (not one row per basket — you build baskets yourself by grouping on InvoiceNo, exactly as in Section 6 of the in-class notebook), with the following columns:

Column	Meaning
InvoiceNo	Transaction / receipt ID. A 6-digit number starting with "C" means a cancellation — exclude these.
StockCode	Product code. Use this (not Description) if you build item embeddings — it's the cleaner, canonical key.

Description	Product name. Strip whitespace before grouping — the same product sometimes appears with trailing spaces.
Quantity	Units purchased. Negative values are returns — exclude these too.
InvoiceDate	Date and time of purchase. Useful if you want to segment by season / day-of-week / hour.
UnitPrice	Price per unit, in GBP.
CustomerID	Customer ID. Can be missing for guest checkouts — association rules work fine without it, since a basket is defined by InvoiceNo, not CustomerID.
Country	Customer's country. This is your natural segmentation column — the in-class notebook used France; you should pick your own.

2. Your Task 🎯

Suggested Investigation Path (guided warm-up)

Before your open-ended investigation, work through these five guided questions from the in-class notebook — they check that your mechanics (cleaning, Apriori, FP-Growth, hand calculation, Item2Vec) are solid before you go explore on your own:

1. Change COUNTRY. Try "Germany" or "United Kingdom" (the full UK market). How does the number of frequent itemsets change? Does `MIN_SUPPORT = 0.03` still make sense, or do you need to lower it?
2. Re-time Apriori vs. FP-Growth on the full UK dataset (Section 11's timing cells). By how many times faster is FP-Growth once the catalog and transaction count get large? Plot runtime vs. `min_support` for both algorithms on the same axes.
3. Hand-calculate a new rule. Using the toy transactions in Section 2, calculate Support, Confidence, and Lift for `{Bread} ⇒ {Milk}` by hand, then verify with `mlxtend` — show your work as you would on a written exam.
4. Tune the Item2Vec model. Try `vector_size=64` and `window=5`. Do the nearest neighbors for your demo product change? Why might a smaller window sometimes give more relevant neighbors for basket data, even though baskets have no real "order"?
5. (Challenge) Build a rule that would not have been findable by Apriori at a reasonable `min_support`, but is findable by comparing Item2Vec embeddings (i.e. two rare, long-tail products that are semantically similar but never individually frequent enough to pass a support threshold). Explain why this is exactly the gap embeddings are meant to fill.

Open-Ended Investigation (the main graded task)

This is where most of your grade comes from. Pick your own angle on the dataset — a country, a customer segment, a product category, a season — and design a market-basket-analysis study around it. A simple, defensible starting framing:

- Pick a market to mine. It does not have to be France. Try a bigger market (United Kingdom, Germany) or compare two markets side by side — do the strongest rules differ by country?
- Justify your `min_support` / `min_confidence` / `min_lift` thresholds. There is no single correct value — explain, with evidence (a plot of itemset count vs. threshold is ideal), why you picked yours.
- Decide what "interesting" means for your investigation. Highest lift? Highest revenue potential (`support × average basket value`)? Rules involving a specific product category you find interesting? State your criterion up front, before you start filtering rules.

- This is the same kind of design decision our Week 6 assignment asked you to make when building an implicit interaction signal from raw fields — here, the design decision is which segment to mine and which threshold/criterion defines an "interesting" rule for that segment. Document your reasoning.

Required Steps

(mirrors our in-class notebook)

1. EDA & cleaning — remove cancelled invoices (InvoiceNo starting with "C"), non-positive Quantity/UnitPrice, and missing Description. Plot the distribution of transactions by country, and the top-selling products for your chosen segment.
2. Build your basket / one-hot transaction matrix — group by InvoiceNo, then encode item presence as 0/1 (not raw quantity), exactly as in Sections 6–7 of class. State clearly which segment (country / date range / customer group) you chose and why.
3. Pick and justify ≥ 2 techniques from: Apriori, FP-Growth, and Item2Vec (embedding-based similarity). Briefly explain, using the comparison table from Section 14 of class, why each technique is a reasonable fit for the segment and scale you chose.
4. Mine frequent itemsets and rules with each chosen technique, using the same basket definition and the same underlying data for all techniques (avoid leakage — decide your cleaning and segment filters once, then reuse them everywhere).
5. Report your top rules / nearest neighbors for at least 3 different "angles" — e.g. three different product categories, three different countries, or three different min_support thresholds — for every technique you built.
6. Evaluate & compare — for Apriori/FP-Growth, report rule count, average Lift of your top rules, and runtime; for Item2Vec, report qualitative neighbor quality and how many long-tail products it can meaningfully cover that rule mining could not. Present your ≥ 2 techniques side by side in one comparison table, exactly like Section 14 of class.
7. Handle the rare-item / long-tail problem explicitly — show at least one product that never reaches your min_support threshold, and demonstrate what (if anything) an embedding-based approach can still tell you about it.
8. Business interpretation — for your chosen segment, state in plain language which rule(s) or technique you'd act on and why, backed by your actual Support/Confidence/Lift numbers.
9. Quantify business impact — pick your single best rule and estimate incremental revenue from one "frequently bought together" prompt, with clearly labeled assumptions (take rate, average price), exactly like the worked example in Section 17 of class.
10. Conclusion — which technique won for your chosen segment, what trade-off did you accept to get there (e.g. interpretability vs. scale, or runtime vs. exhaustiveness), and which technique would you actually ship to production first, and why?

Use this table as a starting skeleton for Step 6 (feel free to add rows/columns your investigation needs):

Technique	# Rules / neighbors found	Avg. Lift (top 10)	Runtime	Coverage of catalog
Apriori				
FP-Growth				
Item2Vec (embeddings)				

⚠ Grading Emphasis — Read This Carefully

Getting one algorithm to run is necessary, but it is not sufficient to score well this week. Most of the marks for this assignment are earned in Steps 6–9 above: how rigorously you compare your ≥ 2 techniques on the same basket data, and how clearly you can explain, in plain business language backed by your own Support/Confidence/Lift numbers, which rule or technique you'd actually deploy for your chosen segment.

A submission that mines one country with Apriori and FP-Growth, but argues its rule-selection criterion rigorously and backs its business recommendation with real numbers, will score higher than a submission that mines five countries with all three techniques but only offers a single vague sentence of comparison at the end.

3. Bonus Points 🌟

- Build a third technique beyond your required two (e.g. add Item2Vec if your two were Apriori + FP-Growth) and extend your comparison table — +bonus
- Compare the same rule-mining pipeline across two different countries side by side (e.g. France vs. Germany) and discuss which patterns are universal vs. market-specific — +bonus
- Build the network-graph visualization from Section 13 of class for your chosen segment's top rules, and identify a "hub" product that appears as the antecedent or consequent of an unusually large number of strong rules — +bonus
- Quantify the long-tail coverage gap: what fraction of unique products in your chosen segment never clear a reasonable min_support threshold, and what fraction of those does your Item2Vec model still produce sensible neighbors for? — +bonus
- Combine rule mining with time: does the strongest rule for your segment change between different months / seasons in the data? — +bonus